

Federate Integration Tech Document

Federate Integration | ITIO | Pawnesh Kumar

Overview

This project is to setup federated Sign In/Signup functionality in PHP Web applications.

By using this feature/functionality user can quickly access your application and you also get a verified user, Whose verification is done by Google, Facebook, LinkedIn or other trusted source which you integrate.

Functionality

Federate Integration gives you authentication link of social media platforms that you have integrated on you web application.

User can click on any of the link that you have integrated, After the click user will be redirected to Authentication page of respective social media there, The user have to enter his social media credentials and then have to click on login, Then the social media algorithm verify the credentials and give a response accordingly.

If the credentials matches it give public user details as a response. Otherwise, Error as a response.

Technologies Used

- PHP
 - It's a programming language used to create dynamic web applications.
- MySQL
 - MySQL is an open-source relational database management system.
- Weaknesses
 - API stands for Application Programming Interface. In the context of APIs, the word Application refers to any software with a distinct function. Interface can be thought of as a contract of service between two applications. This contract defines how the two communicate with each other using requests and responses.

Google Login

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Environment Setup

To setup the environment you will require to follow the given below steps

❑ Google OAuth 2.0 Client Secret & the Client ID

- Go to the 'Google API Console' and login with you Google account. (<https://console.cloud.google.com>)
- Enter on Create New Project link, It will ask you to enter project name. Simply enter the project name and click on Create.
- Select your created project
- Now go to OAuth consent screen, There will be a Field User Type that have two options 1. Internal 2.External, You need to choose 2nd Option (External) and the click on create.
- Enter App name, Support Email, Developer contact information, and click on save & continue.
- Don't do anything at scop section just click save & continue.
- Add test user and click save and continue.
- Then you will need to create OAuth credentials, To do that click on Credentials from left side bar, Create credentials, OAuth client id.
- Select application type, Enter name, Add authorized redirect URIs and click create.
- After you get the client Id and Client Secret, Copy them and save else where .

❑ Google API PHP Client

- You will need to download Google Api PHP client, You can do it in two ways → By using composer → Manual Download
- By using Composer, You need to run 'composer require google/apiclient:^2.0' command in you project root folder.
- Manually you have to search for google api php client in the browser after that there will be a github link, You need to open it and click on download.zip file option Or you can visit <https://github.com/googleapis/google-api-php-client/releases> this link and then choose the latest version and click on download.zip option.
- After downloading the google API PHP client, You need to extract that downloaded folder and then just simply the extracted folder and past in you project root directory.

Database Setup

❑ Existing Database

- If you have existing database, That already have a user table then you just need to add few columns in it.

→ ALTER TABLE 'You user table name ' ADD oauth_provider enum('facebook','google','linkdin')

→ ALTER TABLE 'You user table name 'ADD oauth_uid varchar(50)

→ ALTER TABLE 'You user table name ' ADD modified datetime

You can use above sql quarry to add required columns to you database table.

❑ New Database

```
CREATE TABLE `users_tbl_name` (  
  `id` int(11) NOT NULL,  
  `name` varchar(50) NOT NULL,  
  `first_name` varchar(30) NOT NULL,  
  `last_name` varchar(30) NOT NULL,  
  `email` varchar(50) NOT NULL,  
  `oauth_provider` enum('facebook','google','linkedin'),  
  `oauth_uid` varchar(50)  
  `modified` datetime  
  `profile_image` text NOT NULL ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4  
  COLLATE=utf8mb4_unicode_ci;
```

To create new database you can use above quarry, It will generate a user table along with all necessary columns.

Oauth.php

```
<?php
```

```
require_once 'google-Oauth-api/vendor/autoload.php'; // Adjust the path accordingly
```

```
require_once 'Googleuser.class.php';
```

```
$client = new Google_Client();
```

```
$client->setClientId('Your Client Id');
```

```
$client->setClientSecret('Your Client Secret');
```

```
$client->setRedirectUri('Your Redirect URI');//Redirect URI must match with Console added URI
```

```
$client->addScope('email');
```

```
$client->addScope('profile');
```

```
if (isset($_GET['code'])) {  
  
    $token = $client->fetchAccessTokenWithAuthCode($_GET['code']);  
  
    $client->setAccessToken($token);  
  
    $_SESSION['access_token'] = $token;  
  
    $google_oauth = new Google_Service_Oauth2($client);  
  
    $google_account_info = $google_oauth->userinfo->get();  
  
    $google_data['username'] = $google_account_info->given_name.$google_account_info->id;  
  
    $google_data['first_name'] = $google_account_info->given_name;  
  
    $google_data['last_name'] = $google_account_info->family_name;  
  
    $google_data['full_name'] = $google_account_info->name;  
  
    $google_data['email'] = $google_account_info->email;  
  
    $google_data['oauth_provider'] = "google";  
  
    $google_data['oauth_uid'] = $google_account_info->id;
```

```
$googleUser = new Googleuser();

$userData = $googleUser->checkUser($google_data);

if(!empty($userData)){

    $_SESSION['userData'] = $userData;

    get_member_details($userData['account_no']);

    $_SESSION['s_account_no'] = $userData['account_no'];

    $_SESSION['s_login'] = date("Y-m-d H:i:s");

    $_SESSION['s_username'] = $userData['username'];

    $_SESSION['s_status'] = $userData['status'];

    header("location:company-details.php");exit;

}

else{

    $output = '<h3 style="color:red">Some problem occurred, please try again.</h3>';

}
```

```
else {

    $auth_url = $client->createAuthUrl();

    $googleOutput = "<a href='\"' . $auth_url . \"'\" class='btn btn-outline-danger'>

        <svg xmlns='http://www.w3.org/2000/svg' width='16' height='16'

            fill='currentColor' class='bi bi-google' viewBox='0 0 16 16'>

                <path d='M15.545 6.558a9.4 9.4 0 0 1 .139 1.626c0 2.434-.87 4.492-2.384 5.885h.002C11.978 15.292 10.158 16 8
16A8 8 0 1 1 8 0a7.7 7.7 0 0 1 5.352 2.082l-2.284 2.284A4.35 4.35 0 0 0 8 3.166c-2.087 0-3.86 1.408-4.492 3.304a4.8 4.8 0 0
0 0 3.063h.003c.635 1.893 2.405 3.301 4.492 3.301 1.078 0 2.004-.276 2.722-.764h-.003a3.7 3.7 0 0 0 1.599-2.431H8v-
3.08z' />

        </svg>

        <span style='margin:5px'>

            Google

        </span>

    </a>";

}
```

Google.class.php

```
<?php
private $dbHost    = 'localhost';
private $dbUsername = 'root';
private $dbPassword = '';
private $dbName    = 'bof bank_new-db';
private $userTbl   = 'tbl_account';
function __construct(){
    if(!isset($this->db)){
        // Connect to the database
        $conn = new mysqli($this->dbHost, $this->dbUsername, $this->dbPassword, $this->dbName);
        if($conn->connect_error){
            die("Failed to connect with MySQL: " . $conn->connect_error);
        }else{
            $this->db = $conn;
        }
    }
}
function checkUser($data = array()){
    if(!empty($data)){
        // Check whether the user already exists in the database
        $checkQuery = "SELECT * FROM ".$this->userTbl." WHERE oauth_provider = '".$data['oauth_provider']."' AND oauth_uid = 
        '".$data['oauth_uid']."'";
        $checkResult = $this->db->query($checkQuery);
        if($checkResult->num_rows > 0){
        }
```

```
else{
    // Add created time to the data array
    if(!array_key_exists('add_date',$data)){
        $data['add_date'] = date("Y-m-d H:i:s");
    }
    if(!array_key_exists('status',$data)){
        $data['status'] = "New";
    }
    if(!array_key_exists('password',$data)){
        $chars = "abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789";
        $generate_pass = substr( str_shuffle( $chars ), 0, 8 );
        $password=hash_f($generate_pass);
        $data['password'] = $password;
    }
    // Prepare column and value format
    $columns = $values = "";
    $i = 0;
    foreach($data as $key=>$val){
        $pre = ($i > 0)?', ':'';
        $columns .= $pre.$key;
        $values .= $pre.""". $this->db->real_escape_string($val).""";
        $i++;
    }
}
```

```
// Insert user data in the database

$query = "INSERT INTO ".$this->userTbl." (". $columns.") VALUES (". $values.")";

$insert = $this->db->query($query);

    // Send Mail Function //

$template="SIGNUP-TO-MEMBER";

$postvar['fullname']=$data['name'];

$postvar['sitename']=$_SESSION['host_name'];

$postvar['username']=$data['username'];

$postvar['password']=$generate_pass;

$postvar['email']= $data['email'];

sent_template_mail($template,$postvar);

    // End Send Mail function //

}


// Get user data from the database

$result = $this->db->query($checkQuery);

$userData = $result->fetch_assoc();

}

// Return user data

return !empty($userData)?$userData:false;

}

}
```

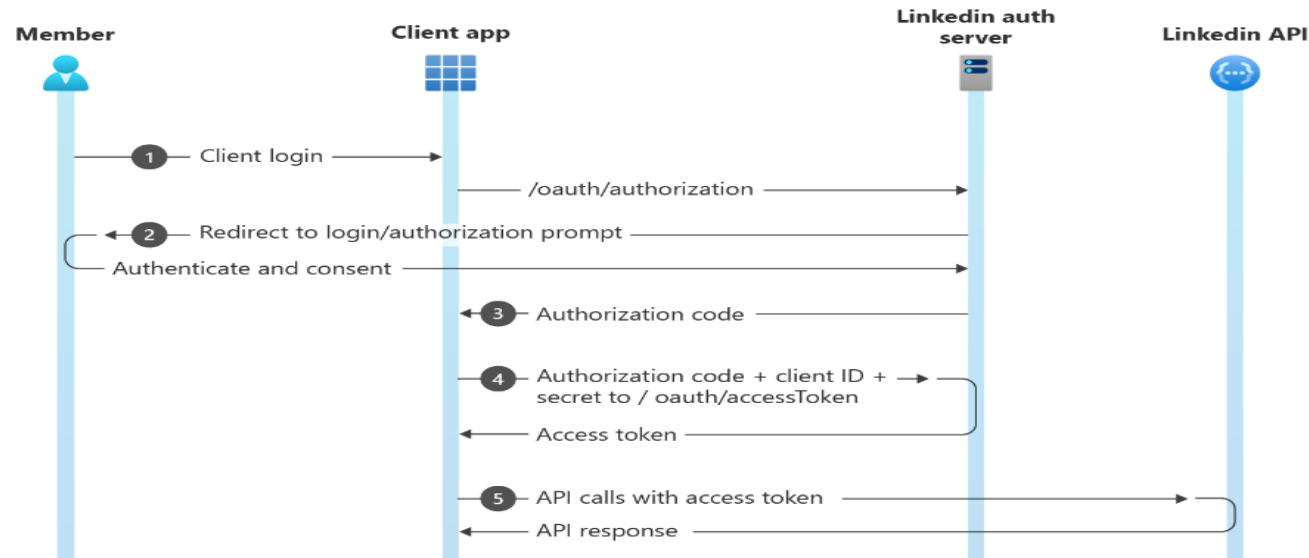
LinkedIn Login

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Environment Setup

To setup the environment you need to follow the below mentioned steps, You also need to have understanding of OAuth 2.0

❑ OAuth 2.0 3 legged OAuth



❑ LinkedIn OAuth 2.0 Client Id & Client Secret

- To obtain client id and client secret you need to create an account on LinkedIn Developers Portal. You can simply search LinkedIn Developers Account or Click on <https://developer.linkedin.com/>
- Click on create new app button to create a new linkedIn app.
- Provide the information about your app and click create app to submit the app registration form.
 - **App name** - Name of the Application.
 - **Company** - Name of your company.
 - **App Description** – Work flow of the application.
 - **App logo** – Logo to Display on OAuth dialog.
 - **Privacy Policy URL** – URL of the privacy policy page.
 - **Business email** – Your business Email.
 - **Product** – Select the product that you would like to add to your app.
- On successful app creation, the page will be redirected to the App settings screen.
- Switch to the Auth tab » Scroll down to OAuth 2.0 settings section.
- Specify the redirect URL in the Redirect URLs fields and Update the App settings.
- In the Application credentials section, you will see the Client ID and Client Secret of your LinkedIn App. The App ID and App secret need to be specified in the script at the time of the LinkedIn API call.

❑ LinkedIn API PHP Client

→ There will be no need to insert any external file for LinkedIn API Client everything is included in the source code.

Database Setup

❑ Existing Database

If you have an existing database, that already has a user table then you just need to add a few columns to it.

→ ALTER TABLE 'Your user table name' ADD oauth_provider enum('facebook','google','linkedin')

→ ALTER TABLE 'Your user table name' ADD oauth_uid varchar(50)

→ ALTER TABLE 'Your user table name' ADD modified datetime

You can use the above SQL queries to add required columns to your database table.

❑ New Database

```
CREATE TABLE `users_tbl_name` (  
  `id` int(11) NOT NULL,  
  `name` varchar(50) NOT NULL,  
  `first_name` varchar(30) NOT NULL,  
  `last_name` varchar(30) NOT NULL,  
  `email` varchar(50) NOT NULL,  
  `oauth_provider` enum('facebook','google','linkedin'),  
  `oauth_uid` varchar(50)  
  `modified` datetime  
  `profile_image` text NOT NULL ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4  
  COLLATE=utf8mb4_unicode_ci;
```

To create new database you can use above quarry, It will generate a user table along with all necessary columns.

File Structure

- Root_directory
 - linkedin_o_auth.php
 - Linkedin.class.php

linkedin_o_auth.php

```
<?php
// LinkedIn App credentials
require_once 'Linkedinuser.class.php';

$client_id = Your Client Id;
$client_secret = 'Your Client Secret';
$redirect_uri = 'Your redirect URI';
$state = 'foobar'; // You can use this to protect against CSRF attacks
$scope = 'openid profile email'; // LinkedIn API Permissions
// Step 1: User clicked on the "Login with LinkedIn" link
if (isset($_GET['code'])) {
    // Step 2: LinkedIn redirects back to your site with the authorization code
    // Step 3: Exchange authorization code for access token
    $code = $_GET['code'];
    $token_url = 'https://www.linkedin.com/oauth/v2/accessToken';
    $token_params = [
        'grant_type' => 'authorization_code',
        'code' => $code,
        'redirect_uri' => $redirect_uri,
        'client_id' => $client_id,
        'client_secret' => $client_secret,
    ];
```

```
$token_response = http_post($token_url, $token_params);
$token_data = json_decode($token_response, true);
// Check if access token is obtained successfully
if (isset($token_data['access_token'])) {
    $access_token = $token_data['access_token'];
    // Step 4: Use access token to fetch user information
    $profile_url = 'https://api.linkedin.com/v2/userinfo';
    $profile_headers = [
        'Authorization: Bearer '.$access_token,
        'Connection: Keep-Alive',
    ];
    $profile_response = http_get($profile_url, $profile_headers);
    $profile_data = json_decode($profile_response, true);
    if (array_key_exists("status", $profile_data)){
    }else{
        // Check user
        $lduserData['first_name'] = $profile_data['given_name'];
        $lduserData['last_name'] = $profile_data['family_name'];
        $lduserData['full_name'] = $profile_data['name'];
        $lduserData['email'] = $profile_data['email'];
        $lduserData['username'] = $profile_data['given_name'].$profile_data['sub'];
        $lduserData['oauth_provider'] = "linkedin";
        $lduserData['oauth_uid'] = $profile_data['sub'];
        $linkedinuser = new Linkedinuser();

        $userData = $linkedinuser->checkUser($lduserData);
        // Storing user data in the session
        $_SESSION['userData'] = $userData;
```

```
$_SESSION['userData'] = $userData;
// LD Auth Logic
if(!empty($userData)){
    get_member_details($userData['account_no']);
    $_SESSION["s_account_no"] = $userData['account_no'];
    $_SESSION["s_login"] = date("Y-m-d H:i:s");
    $_SESSION["s_username"] = $userData['username'];
    $_SESSION["s_status"] = $userData['status'];
    header("location:company-details.php");exit;
}else{
    $output = '<h3 style="color:red">Some problem occurred, please try again.</h3>';
}
}
} else {
    echo 'Error obtaining access token.';
}} else {
// Step 0: User clicks the "Login with LinkedIn" link
$auth_url = 'https://www.linkedin.com/oauth/v2/authorization?';
$auth_url .= 'response_type=code';
$auth_url .= '&client_id=' . $client_id;
$auth_url .= '&redirect_uri=' . urlencode($redirect_uri);
$auth_url .= '&state=' . $state;
$auth_url .= '&scope=' . urlencode($scope);
$linkdOutput = '<a class="btn btn-outline-primary ld-btn" href="'. $auth_url. "'>
```

```

<svg xmlns="http://www.w3.org/2000/svg" width="16" height="16" fill="currentColor" class="bi bi-linkedin" viewBox="0 0 16 16">
  <path d="M0 1.146C0 .513.526 0 1.175 0h13.65C15.474 0 16 .513 16 1.146v13.708c0 .633-.526 1.146-1.175 1.146H1.175C.526 16 0 15.487 0 14.854zm4.943 12.248V6.169H2.542v7.225zm-1.2-8.212c.837 0 1.358-.554 1.358-1.248-.015-.709-.52-1.248-1.342-1.248S2.4 3.226 2.4 3.934c0 .694.521 1.248 1.327 1.248zm4.908 8.212V9.359c0-.216.016-.432.08-.586.173-.431.568-.878 1.232-.878.869 0 1.216.662 1.216 1.634v3.865h2.401V9.25c0-.222-1.184-3.252-2.764-3.252-1.274 0-1.845.7-2.165 1.193v.025h-.016l.016-.025V6.169h-2.4c.03.678 0 7.225 0 7.225z"/>
</svg>
<span style="margin:5px">
  Linkedin
</span>
</a>';
}

```

// Helper function for making HTTP POST requests

```

function http_post($url, $params) {
    $ch = curl_init($url);
    curl_setopt($ch, CURLOPT_POST, 1);
    curl_setopt($ch, CURLOPT_POSTFIELDS, http_build_query($params));
    curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);
    $response = curl_exec($ch);
    curl_close($ch);
    return $response;
}

```

// Helper function for making HTTP GET requests

```

function http_get($url, $headers) {
    $ch = curl_init($url);
    curl_setopt($ch, CURLOPT_HTTPHEADER, $headers);
    curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);
    $response = curl_exec($ch);
    curl_close($ch);
    return $response;
}
?>

```

Linkedin.class.php

```
<?php
class Linkedin {
private $dbHost    = 'localhost';
    private $dbUsername = 'root';
    private $dbPassword = '';
    private $dbName    = 'bof bank_new-db';
    private $userTbl    = 'tbl_account';
    function __construct(){
        if(!isset($this->db)){
            // Connect to the database
            $conn = new mysqli($this->dbHost, $this->dbUsername, $this->dbPassword, $this->dbName);
            if($conn->connect_error){
                die("Failed to connect with MySQL: " . $conn->connect_error);
            }else{
                $this->db = $conn;
            }
        }
    }
    function checkUser($data = array()){
        if(!empty($data)){
            // Check whether the user already exists in the database
            $checkQuery = "SELECT * FROM ".$this->userTbl." WHERE oauth_provider = '".$data['oauth_provider']."' AND oauth_uid = 
            '".$data['oauth_uid']."'";
            $checkResult = $this->db->query($checkQuery);
            if($checkResult->num_rows > 0){

            }
        }
    }
}
```

```

else{
    // Add created time to the data array
    if(!array_key_exists('add_date',$data)){
        $data['add_date'] = date("Y-m-d H:i:s");
    }
    if(!array_key_exists('status',$data)){
        $data['status'] = "New";
    }
    if(!array_key_exists('password',$data)){
        $chars = "abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789";
        $generate_pass = substr( str_shuffle( $chars ), 0, 8 );
        $password=hash_f($generate_pass);
        $data['password'] = $password;
    }
    $columns = $values = "";
    $i = 0;
    foreach($data as $key=>$val){
        $pre = ($i > 0)?', ':'';
        $columns .= $pre.$key;
        $values .= $pre."".$this->db->real_escape_string($val)."";
        $i++;
    }
    $query = "INSERT INTO ".$this->userTbl." ( ".$columns." ) VALUES ( ".$values." )";
    $insert = $this->db->query($query);
    if($insert){
        // Send Mail Function //
        $template="SIGNUP-TO-MEMBER";
        $postvar['fullname']=$data['name'];
        $postvar['sitename']=$_SESSION['host_name'];
        $postvar['username']=$data['username'];
        $postvar['password']=$generate_pass;
        $postvar['email']= $data['email'];
        sent_template_mail($template,$postvar);
        // End Send Mail function //
    }
}

```

```
// Get user data from the database

    $result = $this->db->query($checkQuery);

    $userData = $result->fetch_assoc();

}

// Return user data

return !empty($userData)?$userData:false;

}

}
```

Facebook Login

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Environment Setup

To setup environment for Facebook login you need to follow the following steps

❑ Facebook OAuth 2.0 Client Id & Secrete

- To obtain Client Id & Secrete key, You need to go to Meta Developer Account and login with your facebook credentials
- After login Click on My app option at the top right corner of navbar, Then click on Add new app
- Enter the display name and contact email, Then click on create app button after clicking you will be redirected to the App Dashboard.
- Navigate to the settings → Basic Settings
 - Specify the app domains and select the Category of you App, Click the save changes button.
- Click on the Add product option
 - Select Facebook login product.
 - Select Web as the App platform.
 - Enter the Site URL and Save.
- Navigate to Facebook login → settings page.
 - In the Valid OAuth Redirect URIs field, enter the Redirect URL.
 - Click the Save Changes.

- Go to the Settings → settings page
 - Basic page, note the App ID and App Secret. This App ID and App secret allow you to access the Facebook APIs.
- To retrieve the user's Facebook timeline link and gender, you need to submit a request for user_link and user_gender permissions.
 - Go to the App Review » Permissions and Features page.
 - o Once the review process is completed and approved by the Facebook, you will be able to get the user profile link and gender from the Facebook Graph API.

Note that: The App ID and App secret need to be specified in the script at the time of Facebook API call. Also, the Valid OAuth Redirect URIs must be matched with the Redirect URL that specified in the script.

Database Setup

❑ Existing Database

- If you have existing database, That already have a user table then you just need to add few columns in it.

→ ALTER TABLE 'You user table name ' ADD oauth_provider enum('facebook','google','linkdin')

→ ALTER TABLE 'You user table name 'ADD oauth_uid varchar(50)

→ ALTER TABLE 'You user table name ' ADD modified datetime

You can use above sql quarry to add required columns to you database table.

❑ New Database

```
CREATE TABLE `users_tbl_name` (  
  `id` int(11) NOT NULL,  
  `name` varchar(50) NOT NULL,  
  `first_name` varchar(30) NOT NULL,  
  `last_name` varchar(30) NOT NULL,  
  `email` varchar(50) NOT NULL,  
  `oauth_provider` enum('facebook','google','linkedin'),  
  `oauth_uid` varchar(50)  
  `modified` datetime  
  `profile_image` text NOT NULL ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4  
  COLLATE=utf8mb4_unicode_ci;
```

To create new database you can use above quarry, It will generate a user table along with all necessary columns.

File Structure

- Root_directory
 - facebook_api_config.php
 - facebook_login.php
 - facebook_logout.php
 - User.class.php

Facebook_api_config.php

```
<?php
// Facebook API configuration

//Initialized localhost as false
$localhosts=false;
$con_data['localhosts']=false;

if($_SERVER["HTTP_HOST"]=="localhost" || isset($_SESSION["http_host_loc"])){ //check localhost or not, if yes then define localhost in data
and set true
    $localhosts=true;
    $data['localhosts']=true;
}
$base_url = "https://secure.bofbank.com";
$redirectURL = "https://secure.bofbank.com/auth_login";
if($data['localhosts']==true){
$base_url = "http://localhost/bank-portal/";
$redirectURL = "http://localhost/bank-portal/auth_login";
}
$base_urls = htmlspecialchars($base_url);
$redirectURLs = htmlspecialchars($redirectURL);
define('FB_APP_ID', '769521561676065');
define('FB_APP_SECRET', 'a4d4c0dc24f131e152dabd20863b0b7a');
define('FB_REDIRECT_URL', $redirectURLs);
define('base_url', $base_urls);
```

```
// Start session
if(!session_id()){
    session_start();
}

// Include the autoloader provided in the SDK
require_once "assets/facebook-sdk/autoload.php";

// Include required libraries
use Facebook\Facebook;
use Facebook\Exceptions\FacebookResponseException;
use Facebook\Exceptions\FacebookSDKException;

// Call Facebook API
$f = new Facebook(array(
    'app_id' => FB_APP_ID,
    'app_secret' => FB_APP_SECRET,
    'default_graph_version' => 'v3.2',
));
// Get redirect login helper
$helper = $f->getRedirectLoginHelper();

// Try to get access token
try {
    if(isset($_SESSION['facebook_access_token'])){
        $accessToken = $_SESSION['facebook_access_token'];
    }else{
        $accessToken = $helper->getAccessToken();
    }
} catch(FacebookResponseException $e) {
    echo 'Graph returned an error: ' . $e->getMessage();
    exit;
} catch(FacebookSDKException $e) {
    echo 'Facebook SDK returned an error: ' . $e->getMessage();
    exit;
}
?>
```

facebook_login.php

```
<?php
// Include configuration file
require_once 'facebook_api_config.php';

// Include user class
require_once 'User.class.php';

if(isset($accessToken)){
    if(isset($_SESSION['facebook_access_token'])){
        $fb->setDefaultAccessToken($_SESSION['facebook_access_token']);
    }else{
        // Put short-lived access token in session
        $_SESSION['facebook_access_token'] = (string) $accessToken;

        // OAuth 2.0 client handler helps to manage access tokens
        $oAuth2Client = $fb->getOAuth2Client();
        // Exchanges a short-lived access token for a long-lived one
        $longLivedAccessToken = $oAuth2Client->getLongLivedAccessToken($_SESSION['facebook_access_token']);
        $_SESSION['facebook_access_token'] = (string) $longLivedAccessToken;

        // Set default access token to be used in script
        $fb->setDefaultAccessToken($_SESSION['facebook_access_token']);
    }
}
```

```
// Redirect the user back to the same page if url has "code" parameter in query string
if(isset($_GET['code'])){
    header('Location: http://localhost/bank-portal/');
}

// Getting user's profile info from Facebook
try {
    $graphResponse = $fb->get('/me?fields=name,first_name,last_name,email,link,picture');
    $fbUser = $graphResponse->getGraphUser();
} catch(FacebookResponseException $e) {
    echo 'Graph returned an error: ' . $e->getMessage();
    session_destroy();
    // Redirect user back to app login page
    header("Location: ./");
    exit;
} catch(FacebookSDKException $e) {
    echo 'Facebook SDK returned an error: ' . $e->getMessage();
    exit;
}

// Initialize User class
$user = new User();
// Getting user's profile data
$fbUserData = array();
if(!empty($fbUser['id'])){
    $fbUserData['oauth_uid'] = $fbUser['id'];
}
if(!empty($fbUser['name'])){
    $fbUserData['full_name'] = $fbUser['name'];
}
if(!empty($fbUser['first_name'])){
    $fbUserData['first_name'] = $fbUser['first_name'];
}
```

```
if(!empty($fbUser['last_name'])){
    $fbUserData['last_name'] = $fbUser['last_name'];
}
if(!empty($fbUser['email'])){
    $fbUserData['email'] = $fbUser['email'];
}
if(!empty($fbUser['gender'])){
    $fbUserData['gender'] = $fbUser['gender'];
}
if(!empty($fbUser['birthday'])){
    $fbUserData['birth_date'] = $fbUser['birthday'];
}
$fbUserData['oauth_provider'] = 'facebook';
$fbUserData['username'] = $fbUser['first_name'].$fbUser['id'];
// Insert or update user data to the database
$userData = $user->checkUser($fbUserData);
// Storing user data in the session
$_SESSION['userData'] = $userData;
// Get logout url
$logoutURL = $helper->getLogoutUrl($accessToken,base_url.'fb_logout.php');
// FB Auth Logic
if(!empty($userData)){
    get_member_details($userData['account_no']);
    $_SESSION["s_account_no"] = $userData['account_no'];
    $_SESSION["s_login"] = date("Y-m-d H:i:s");
    $_SESSION["s_username"] = $userData['username'];
    $_SESSION["s_status"] = $userData['status'];
    header("location:company-details.php");exit;
}else{
    $output = '<h3 style="color:red">Some problem occurred, please try again.</h3>';
}
```

```

}else{
    // Get login url

    $permissions = ['email']; // Optional permissions

    $loginURL = $helper->getLoginUrl(FB_REDIRECT_URL,$permissions);

    // Render Facebook login button

    $output = '<a class="btn btn-outline-primary fb-btn" href="'.htmlspecialchars($loginURL)."'>
        <svg xmlns="http://www.w3.org/2000/svg" width="16" height="16" fill="currentColor" class="bi bi-facebook" viewBox="0 0 16
16">

            <path d="M16 8.049c0-4.446-3.582-8.05-8.05-8.05C3.58 0-.002 3.603-.002 8.05c0 4.017 2.926 7.347 6.75 7.951v-5.625h-
2.03V8.05H6.75V6.275c0-2.017 1.195-3.131 3.022-3.131.876 0 1.791.157 1.791.157v1.98h-1.009c-.993 0-1.303.621-1.303 1.258v1.51h2.218l-
.354 2.326H9.25V16c3.824-.604 6.75-3.934 6.75-7.951"/>

        </svg>

        <span style="margin:5px">

            Facebook

        </span>

    </a>';
}

?>

```

User.class.php

```
<?php
class User {
    private $dbHost    = 'localhost';
    private $dbUsername = 'root';
    private $dbPassword = '';
    private $dbName     = 'bof bank_new-db';
    private $userTbl    = 'tbl_account';
    function __construct(){
        if(!isset($this->db)){
            // Connect to the database
            $conn = new mysqli($this->dbHost, $this->dbUsername, $this->dbPassword, $this->dbName);
            if($conn->connect_error){
                die("Failed to connect with MySQL: " . $conn->connect_error);
            }else{
                $this->db = $conn;
            }
        }
    }
    function checkUser($data = array()){
        if(!empty($data)){
            // Check whether the user already exists in the database
            $checkQuery = "SELECT * FROM ".$this->userTbl." WHERE oauth_provider = '".$data['oauth_provider']."' AND oauth_uid = '".$data['oauth_uid']."'";
            $checkResult = $this->db->query($checkQuery);

            // Add modified time to the data array
            if(!array_key_exists('modified',$data)){
                $data['modified'] = date("Y-m-d H:i:s");
            }

            if($checkResult->num_rows > 0){
```

```

}else{
    // Add created time to the data array
    if(!array_key_exists('add_date',$data)){
        $data['add_date'] = date("Y-m-d H:i:s");
    }
    if(!array_key_exists('status',$data)){
        $data['status'] ="New";
    }
    if(!array_key_exists('password',$data)){
        $chars = "abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789";
        $generate_pass = substr( str_shuffle( $chars ), 0, 8 );
        $password=hash_f($generate_pass);
        $data['password'] = $password;
    }
    // Prepare column and value format
    $columns = $values = '';
    $i = 0;
    foreach($data as $key=>$val){
        $pre = ($i > 0)?', ':'';
        $columns .= $pre.$key;
        $values .= $pre."".$this->db->real_escape_string($val)."";
        $i++;
    }
    // Insert user data in the database
    $query = "INSERT INTO ".$this->userTbl." ( ".$columns.") VALUES ( ".$values.)";
    $insert = $this->db->query($query);
    if($insert){
        // Send Mail Function //
        $template="SIGNUP-TO-MEMBER";
        $postvar['fullname']=$data['name'];
        $postvar['sitename']=$_SESSION['host_name'];
        $postvar['username']=$data['username'];
        $postvar['password']=$generate_pass;
        $postvar['email']= $data['email'];
        sent_template_mail($template,$postvar);
        // End Send Mail function //
    }
}

```

```
// Get user data from the database

    $result = $this->db->query($checkQuery);

    $userData = $result->fetch_assoc();

}

// Return user data

return !empty($userData)?$userData:false;

}

}
```

facebook_logout.php

```
<?php
// Include configuration file
require_once 'fb_api_config.php';

// Remove access token from session
unset($_SESSION['facebook_access_token']);

// Remove user data from session
unset($_SESSION['userData']);

// Redirect to the homepage
header("Location:index.php");
?>
```

Special Feature

❑ Limit user to have only one account, While login through different social media platform that have same email ids

→ To achieve this you have to make few flags

1. While fetching data from social media API, Must fetch email id of user.
2. Write a check if fetched data have email id or not.
3. In case there is no email id, Unset the access token and redirect the user on Auth login page with error for ex-: "Your social media account doesn't have email id".
4. Write a check if the email that you got from API already exist in your data base if, Yes then write another check if that record have UID of respective provider or not if, Yes then check if email id and UID exist in your record or not if yes, Then give login access to the user otherwise redirect the user on login page with error.
5. In the 4th step , If the first check give false then create new user.
6. In the 4th step, If the second check (Whether UID of respective provider is exist or not) give false, then update that record with provider UID and give login access to the user.

Note: In the next slide I have added the code changes in respective files take reference from there.

Provider OAuth.php

```
$profileData = json_decode($profile_response, true); // Response get  
from Provider API  
if(array_key_exists("email",$profile_data) &&  
!empty($profile_data['email'])) // Checking if email is exist in response  
or not  
  
// If yes then make data array of response  
  
}else{  
$error_message = "Your Social media account doesn't have Email-Id";  
// Redirect to login page with error  
}
```

Provider Class.php

```
if(!empty($data)){
    // Check whether the user already exists in the database
    $checkQuery = "SELECT * FROM ".$this->userTbl." WHERE email = '".$data['email']."' ";
    $checkResult = $this->db->query($checkQuery);

    // Add modified time to the data array
    if(!array_key_exists('modified',$data)){
        $data['modified'] = date("Y-m-d H:i:s");
    }
    // Checking if email Exist!
    if($checkResult->num_rows > 0){
        $checkData = $checkResult->fetch_assoc();
        if(!empty($checkData['facebook_uid']))
        {
            $checkUID= "SELECT * FROM ".$this->userTbl." WHERE email = '".$data['email']."' AND provider_uid = '".$data['provider_uid']."'";
            $checkUIDResult = $this->db->query($checkUID);
            if($checkUIDResult->num_rows > 0)
            {
                //
            }else{
                $_SESSION['msg']="Access Denied, Different Provider ID is registered with us with the same user";
                header("location:auth_login.php");
            }
        }else{
            // Update UID
            $fire_Result=db_query("update ".$this->userTbl." set provider_uid='".$data['provider_uid']."' where user_id=
            '".$checkData['user_id']."'");
        }
    }else{
        // Create new User.
    }
}
```